

Sr.No	Module	Weightage %	Marks Weightage%
1	Module-V	20.00	21.05
2	Module-I	20.00	19.74
3	Module-II	20.00	19.74
4	Module-III	20.00	19.74
5	Module-IV	20.00	19.74

Weightage representation okay : Weightage representation exceeded more than 5% :Weightage representation exceeded more than 10%

CSE601

[ET]

Enroll. No. - _____

END SEMESTER EXAMINATION : April-May, 2026

Data Structures and Algorithm Design

Time : 3.00 Hrs

Maximum Marks : 60

Note: Attempt questions from all sections as directed.

Section - A : Attempt any Four questions out of Five . Each question carries 06 marks.

Q1.

(a)

Write an algorithm to insert an operation in a BST.

(b)

Define Time Complexity and Space Complexity.

(3)

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Q2.

Write Java Code to convert a given Infix Expression to Postfix Expression.

(6)

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Q3.

Explain Graph representation methods: Adjacency Matrix and Adjacency List.

(6)

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Q4.

(a)

Define the 0/1 Knapsack problem statement.

(b)

Explain Doubly Circular Linked List.

(3)

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(3)

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Q5.

Write Java Code for DFS Graph Traversal.

(6)

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Section – B : Attempt any two questions out of three. Each question carries 10 marks.

Q6.

(a)

Explain the concept of a General Tree and its operations.

(b)

Describe a Red-Black tree rotation (Left and Right).

(5)

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(5)

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Q7.

(a)

Develop a Java code snippet for the insert operation in a BST.

(b)

Explain the "All-pair Shortest Path" problem and mention the algorithm used to solve it.

(5)

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(5)

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Q8.

(a)

Explain the process of proving a problem is NP-Complete.

(b)

Discuss the Nolish Notation algorithm with example.

(5)

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(5)

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Section - C : Compulsory question

Q9.

(a)

Describe the structure of Stacks and Queues and provide one real-world application for each.

(b)

Discuss the Divide and Conquer strategy for Merge Sort and analyze its time complexity.

(c)

Write a Java Code snippet for the BSF graph traversal algorithm.

(d)

Compare 0/1 Knapsack (DP) with Fractional Knapsack.

(4)

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(4)

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(4)

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